DO NOW — RECALL

Lesson 131 · Asymmetric Encryption and Public-Key Use

Name: _____ Class: _____ Date: _____

1. What is the principal challenge of symmetric encryption between distant users?

2. What is a function parameter?

3. What is the binary result of 1 + 1 in a full adder with carry-in 0?

4. How many bits are in three bytes?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Securely sharing and managing the secret key.

2. Model answer

An input used by a subroutine.

3. Model answer

Sum 0, carry-out 1.

4. Model answer

24.

By the end, you can…

• Public key can be shared; private key remains secret.

• A message encrypted with recipient public key is decrypted by recipient private key.

• Signature created with private key is verified using corresponding public key.

Vocabulary — know these terms

public key — A key that can be distributed openly.

private key — A matching secret key kept by its owner.

asymmetric encryption — Uses a related public/private key pair.

digital signature — Cryptographic evidence of origin and integrity with private-key signing.

NEW LEARNING

Read, annotate and explain

Generate key pair

Public key can be shared; private key remains secret.

Encrypt to recipient

A message encrypted with recipient public key is decrypted by recipient private key.

Sign and verify

Signature created with private key is verified using corresponding public key.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

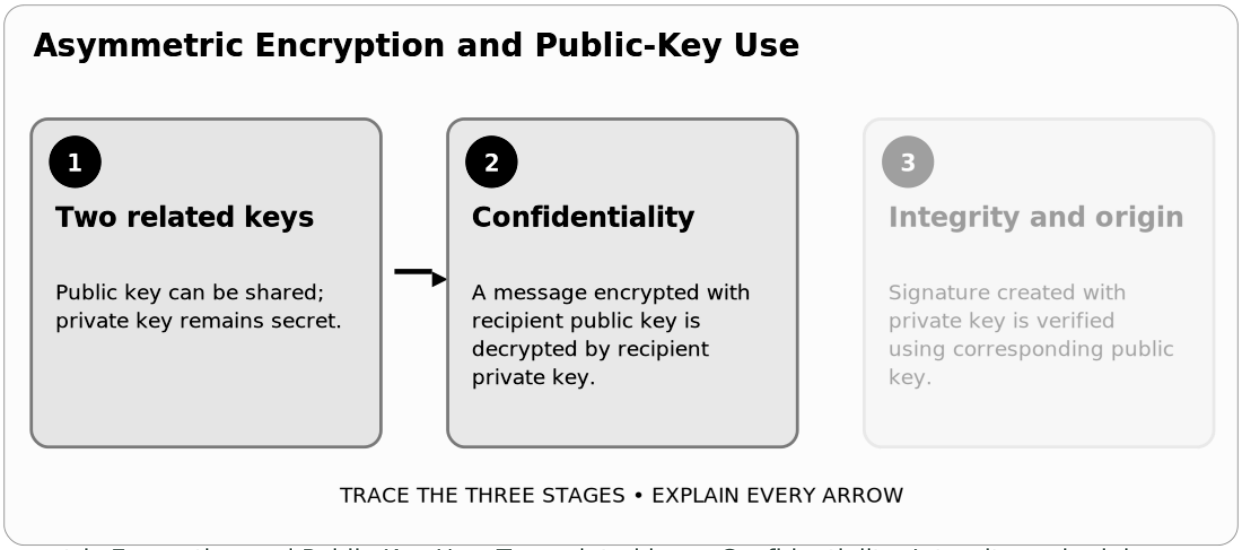
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Asymmetric Encryption and Public-Key Use: Two related keys; Confidentiality; Integrity and origin

New Learning

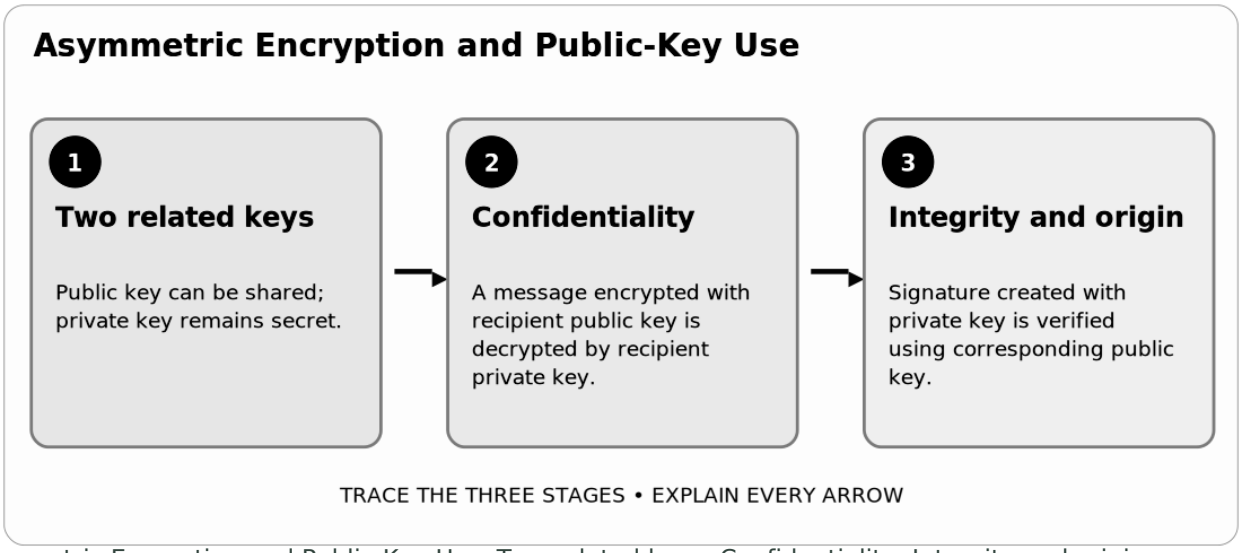


Asymmetric Encryption and Public-Key Use: Two related keys; Confidentiality; Integrity and origin

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Asymmetric Encryption and Public-Key Use: Two related keys; Confidentiality; Integrity and origin

I DO — WORKED EXAMPLE

Follow the worked method

Alice wants to send Bob a private message using public-key encryption. Identify both keys used.

Bob shares public key

Predict the first step from the prompt.

Alice encrypts for Bob

Explain the intermediate result or decision.

Bob uses private key

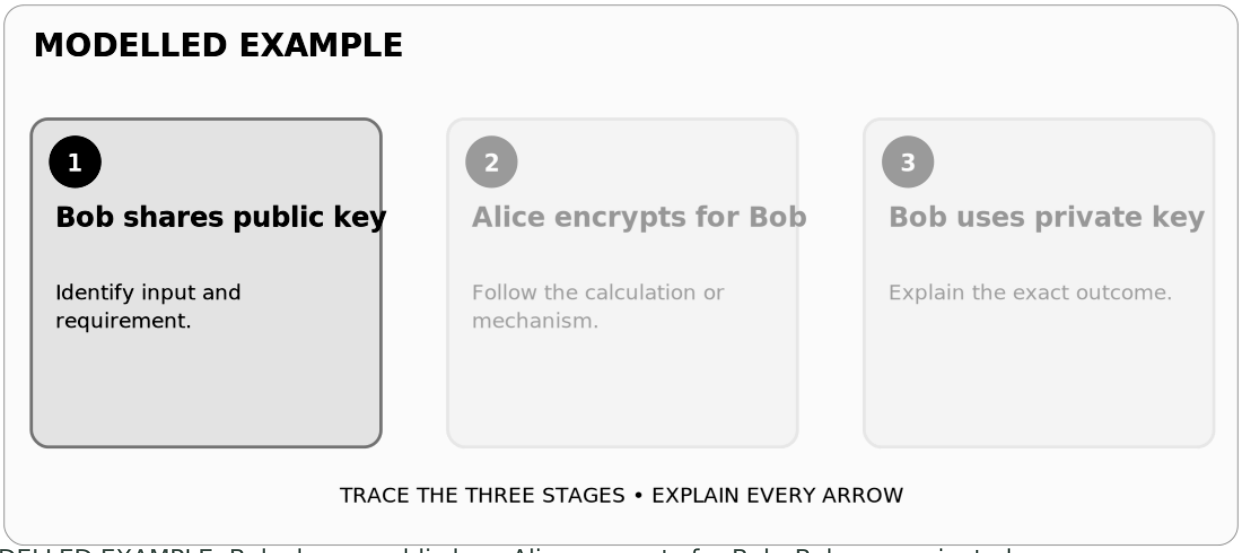
Alice encrypts a suitably sized message or session key using Bob's public key. Bob decrypts using his matching private key. Alice never requires Bob's private key.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Alice wants to send Bob a private message using public-key encryption. Identify both keys used.

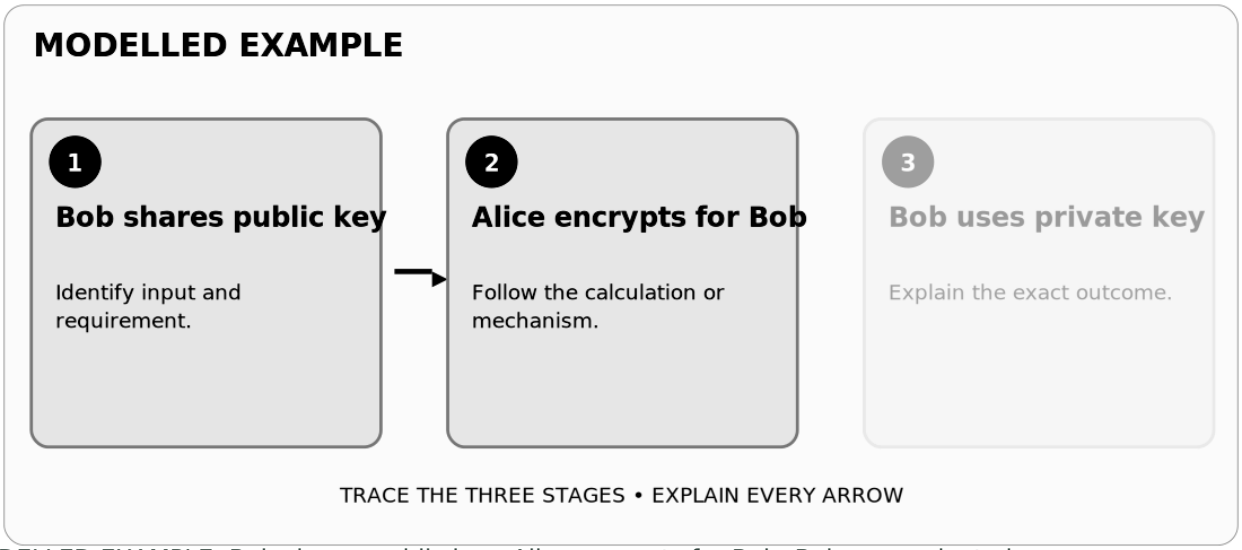
Predict the first step from the prompt.



MODELLED EXAMPLE: Bob shares public key; Alice encrypts for Bob; Bob uses private key

Alice wants to send Bob a private message using public-key encryption. Identify both keys used.

Predict the first step from the prompt.



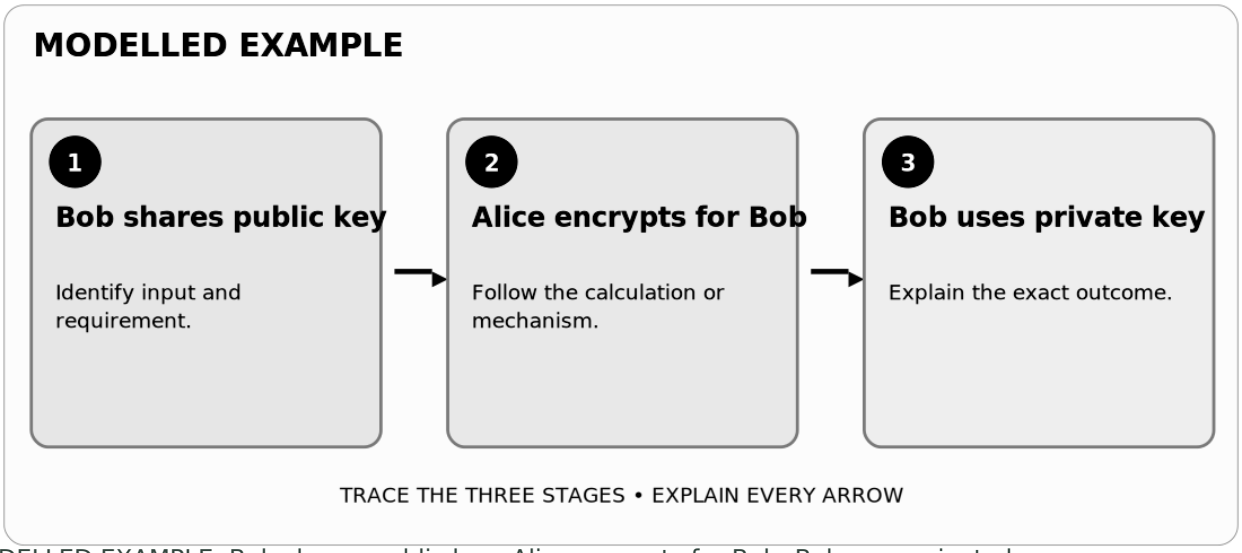
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I Do · original lesson diagrams

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WE DO — GUIDED PRACTICE

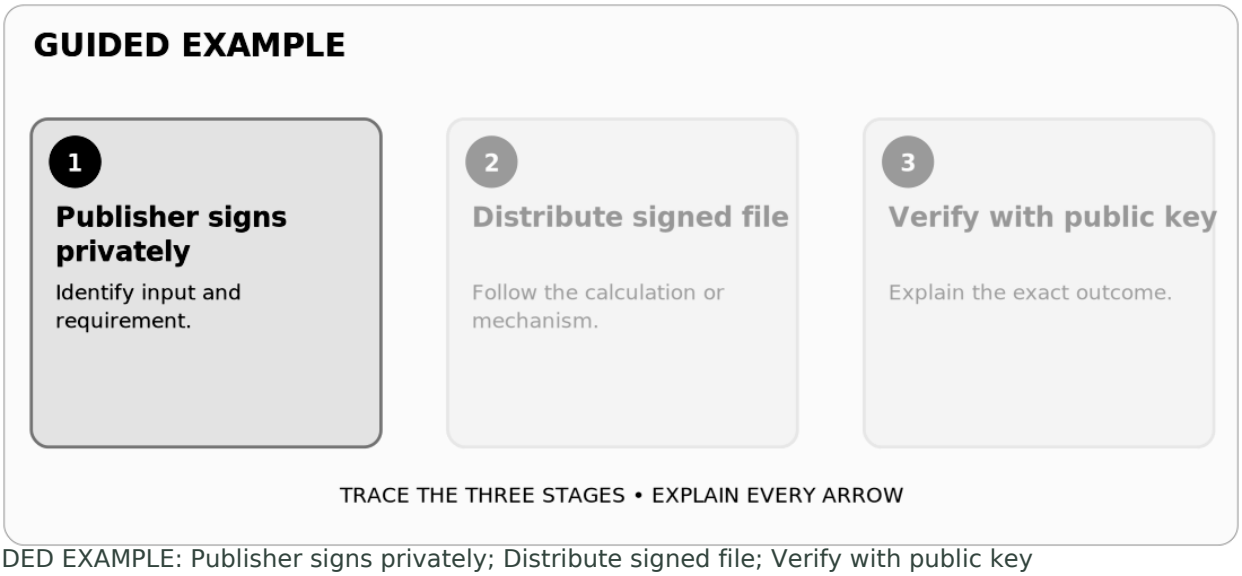
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A website publishes a signed update. Which key verifies its authentic signature?

Predict the first step from the prompt.



GUIDED EXAMPLE: Publisher signs privately; Distribute signed file; Verify with public key

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Publisher signs privately Predict the first step from the prompt.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain the difference between public and private keys and how they support confidentiality. [6]

Question 2

Outline how a digitally signed download is verified. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compare asymmetric and symmetric approaches for sending a large encrypted video file. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A public key must be kept secret for public-key encryption to work.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

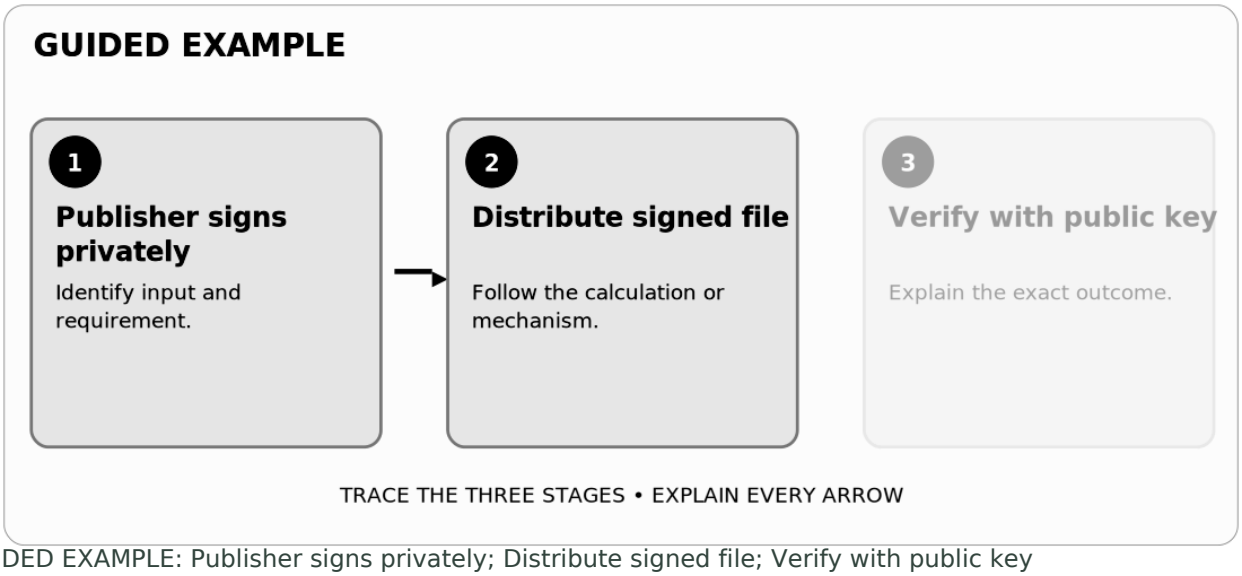
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

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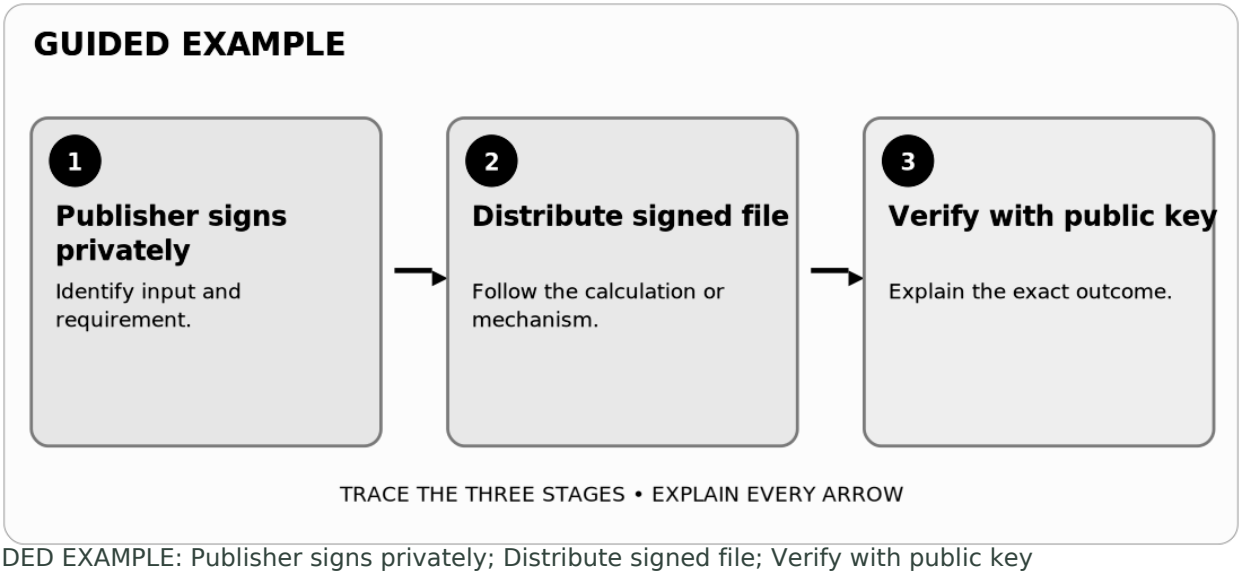
Predict the first step from the prompt.



GUIDED EXAMPLE: Publisher signs privately; Distribute signed file; Verify with public key

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GUIDED EXAMPLE: Publisher signs privately; Distribute signed file; Verify with public key

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Distribute signed file

Explain the intermediate result or decision.

Verify with public key

The publisher creates the digital signature with its private key. Users verify it with the public key obtained through a trusted route; this supports authenticity and integrity, not secrecy of the update.

IMPROVE: Explain one correction to your own reasoning.